

SYSTEM IDENTIFICATION REPORT

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Project Title: Modeling Energy Loss in a Shock Absorber

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1. ABSTRACT

System identification involves developing mathematical models of dynamic systems using observed or assumed data. In this project, a shock absorber is modelled as a mass-spring-damper system to study energy loss characteristics. The damping effect is analyzed to understand how mechanical energy is dissipated over time. Simulation is performed to validate the system behavior.

2. INTRODUCTION

Shock absorbers are essential components in mechanical and automotive systems. They reduce vibrations and improve system stability by dissipating energy.

System identification helps in representing such real-world systems mathematically and predicting their behavior under different conditions.

3. PROBLEM STATEMENT

To develop a mathematical model of a shock absorber and analyze how energy is lost due to damping.

4. OBJECTIVES

- To model the shock absorber as a dynamic system
- To identify system parameters
- To analyze energy dissipation
- To simulate system behavior

5. LITERATURE REVIEW

Previous studies show that shock absorbers can be modeled using second-order differential equations. Numerical methods like Euler and Runge-Kutta are used for simulation. Damping plays a key role in controlling oscillations and energy loss.

6. REAL WORLD SYSTEM

In real-world applications, shock absorbers are used in vehicles to reduce vibrations caused by road irregularities.

When a vehicle encounters a bump:

- The spring stores energy
- The damper dissipates energy as heat
- Oscillations gradually decrease



7. SYSTEM DESCRIPTION

The shock absorber is modelled as a **mass-spring-damper system**:

- Mass (m): Vehicle body
- Spring (k): Stores energy
- Damper (c): Dissipates energy

Input: Road disturbance

Output: Displacement

8. MATHEMATICAL MODELLING

Using Newton's Second Law:

$$m \frac{d^2x}{dt^2} + c \frac{dx}{dt} + kx = 0$$

This represents the motion of the system.

9. DATA / ASSUMPTIONS

Assumed values:

- $m = 250$ kg
- $k = 20000$ N/m
- $c = 1500$ Ns/m

10. PARAMETER IDENTIFICATION

The parameters are selected based on typical vehicle suspension systems. These values determine system response and damping behavior.

11. TRANSFER FUNCTION

Taking Laplace Transform:

$$ms^2X(s) + csX(s) + kX(s) = 0$$

Transfer Function:

$$\frac{x(s)}{F(s)} = \left(\frac{1}{ms^2 + cs + k} \right)$$

12. ENERGY ANALYSIS (DERIVATION)

Total Energy:

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

Multiplying equation of motion by velocity:

$$mx''x' + cx'^2 + kxx' = 0$$

Converting:

$$\frac{d}{dt} \left[\frac{1}{2}mv^2 + \frac{1}{2}kx^2 \right] = -cx^2$$

Final:

$$\frac{dE}{dt} = -cx^2$$

13. METHODOLOGY

1. Develop mathematical model
 2. Apply numerical method
 3. Simulate using Python
 4. Analyze graphs
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14. SIMULATION

The system is simulated using Python to observe displacement and energy decay over time.

15. RESULTS

The simulation of the mass-spring-damper model of the shock absorber provides important insights into the system behavior.

Initially, when the system is disturbed (given an initial displacement), the mass begins to oscillate due to the restoring force of the spring. However, unlike an ideal system, these oscillations do not continue indefinitely. The presence of the damper introduces a resistive force proportional to velocity, which gradually reduces the amplitude of oscillations.

The displacement vs time graph shows a **decaying oscillatory response**, indicating that the system is underdamped. The peaks of the oscillations decrease exponentially with time, demonstrating the effect of energy dissipation.

The energy plot clearly shows that the **total mechanical energy of the system decreases continuously**. At the beginning, the system has maximum energy due to initial displacement. As time progresses:

- Kinetic energy increases and decreases periodically
- Potential energy is exchanged with kinetic energy
- A portion of energy is continuously lost due to damping

This energy loss follows the relation:

- Energy decay is proportional to the square of velocity
- Higher velocity → higher energy loss

Eventually, the system reaches equilibrium where:

- Displacement → 0
- Velocity → 0
- Total energy → minimum (almost zero)

This confirms that the shock absorber effectively removes unwanted vibrations.

16. DISCUSSION

The results obtained from the simulation strongly support the theoretical model of the shock absorber as a mass-spring-damper system.

One of the most important observations is the role of the **damping coefficient (c)**. The value of damping determines how quickly the system loses energy and stabilizes:

- If damping is low → oscillations persist for a longer time
- If damping is high → oscillations decay quickly
- If damping is too high → system becomes overdamped and responds slowly

Thus, an optimal value of damping is required in practical systems to balance comfort and stability.

Another key observation is that **energy loss is not constant**. It depends on velocity:

- At higher velocities → more energy is dissipated
- At lower velocities → energy loss decreases

This matches real-world behavior of hydraulic shock absorbers where fluid resistance increases with speed.

The model used in this project assumes:

- Linear damping
- No external disturbances
- Ideal spring behavior

However, real systems may exhibit:

- Nonlinear damping
- Temperature effects
- Variable fluid viscosity

Despite these simplifications, the model provides a **very good approximation** of real shock absorber behavior and is widely used in engineering analysis.

17. CONCLUSION

In this project, a shock absorber system was successfully modeled using system identification techniques. The system was represented as a mass-spring-damper model, and its behavior was analyzed through mathematical derivation and simulation.

The study clearly demonstrates that damping plays a crucial role in reducing oscillations and stabilizing the system. The derived energy equation shows that the rate of energy loss is proportional to the square of velocity, confirming that energy is continuously dissipated in the form of heat.

Simulation results validate the theoretical model by showing:

- Decaying oscillations over time
- Continuous reduction in system energy
- Stabilization of the system at equilibrium

This confirms that shock absorbers are effective in converting unwanted mechanical energy into thermal energy, thereby improving system performance and comfort.

Overall, the system identification approach proves to be a powerful tool for modeling and analyzing real-world dynamic systems. The findings of this project can be applied to various engineering applications such as vehicle suspension systems, vibration control, and structural stability.

18. REFERENCES

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